Potato - Early blight

RAG record ADV-004 | canonical key: potato_early_blight

Record ID	ADV-004
Crop / crop key	Potato / potato
Condition / disease key	Early blight / early_blight
Main pathogen	Alternaria solani
RAG aliases	potato early blight, Alternaria solani, target spot potato, concentric rings

Farmer quick guide: treatments protect healthy plant tissue and slow disease spread. Dead or badly damaged leaf tissue will not become green again.

How to recognize it

- Brown to black leaf spots often have clear concentric rings, giving a target-board appearance.
- Spots may join together and dry the leaf. Disease is often worse on older or stressed foliage.

Weather or field conditions that favour it

Warm conditions (about 25-30 C), intermittent rain and weak or poorly nourished crops favour early blight.

What to do now

- Use disease-free seed tubers and remove infected crop debris because the fungus can survive in residue and soil.
- Keep the crop well nourished and avoid severe plant stress.
- If disease starts, protect remaining healthy foliage with a locally registered fungicide.

Chemical treatment - only after the disease is confirmed

Chemical / formulation	Rate from cited source	How to use this information	Source
Zineb or captan	0.2% spray (about 2 g/L for a WP formulation)	TNAU reports good control from early spraying, repeated every	[S1]

Chemical / formulation	Rate from cited source	How to use this information	Source
		15-20 days when disease pressure continues.	

Safety and compliance: Use only a product whose current Indian label allows the crop and target disease. Follow the label rate, water volume, re-entry interval and pre-harvest interval. Wear label-required PPE. Do not mix products unless the label or a qualified extension adviser permits it.

Prevention and integrated management

- Plant clean seed tubers.
- Destroy infected debris after harvest.
- Use resistant/tolerant varieties where locally suitable.
- Maintain balanced nutrition and irrigation.

RAG guardrail

The AI should treat the chemical table as source-grounded examples, not permission to invent a different pesticide or dose. If the retrieved source does not match the crop, disease and formulation, return non-chemical advice and request local verification.

Sources used for this record

[S1] Tamil Nadu Agricultural University (TNAU). Potato - Early blight: Alternaria solani.
https://www.agritech.tnau.ac.in/crop_protection/crop_prot_crop%20diseases_veg_potato_2.html